diskrete = bil. bulat
integer

skewed right

skewed left

t ukur

Variabel
- Qualitative → non numerik
  ID, gender, major
- Quantitative → numerik
  GPA, semester, # units

Variabel Nilai UTS	Body temperature	hair color
78	36,2°	black
79	38,1°	brown
100	35°	red
32	37°	black
10		grey

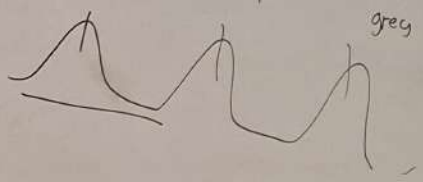
data

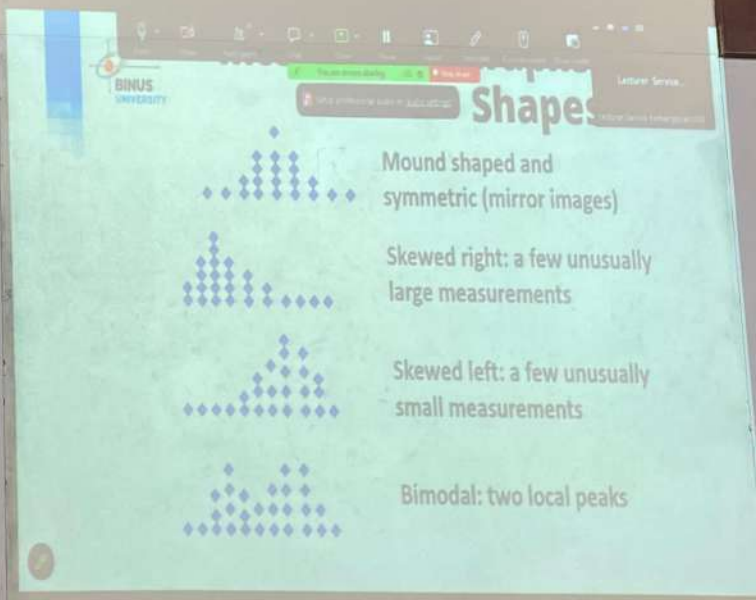
non angka → Gender
Major

angka → ID, GPA, semester, # of units

Gender - 2, 2, 1, 1, 2 (angka)

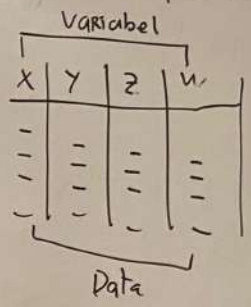
1 = M
2 = F





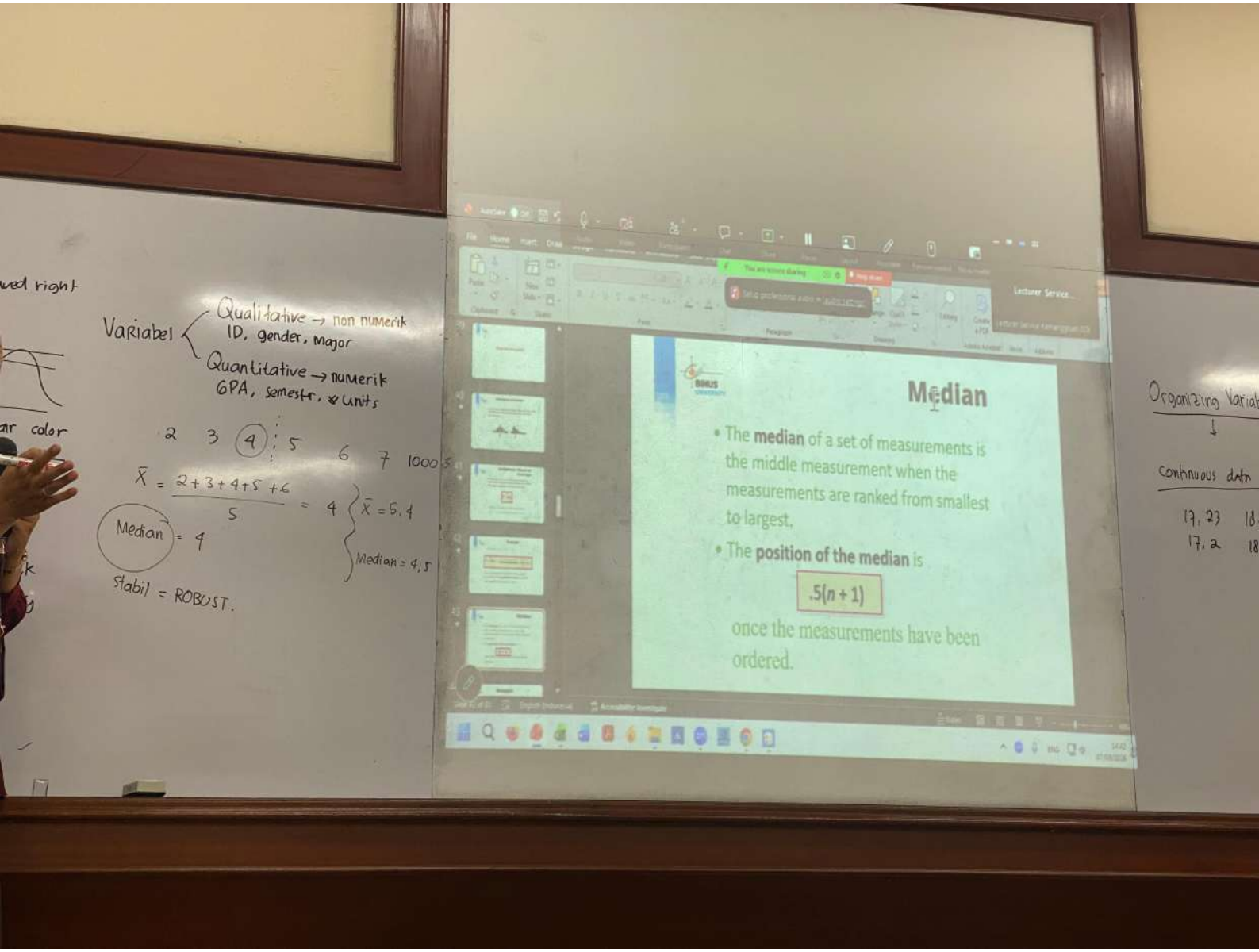
Statistik

- Variabel : karakteristik / alat ukur
- Data : hasil pengukuran

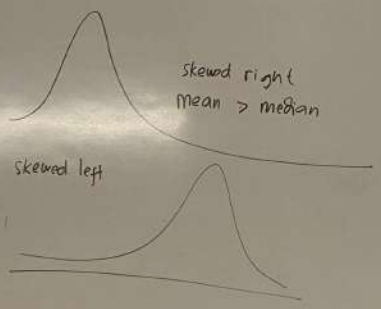


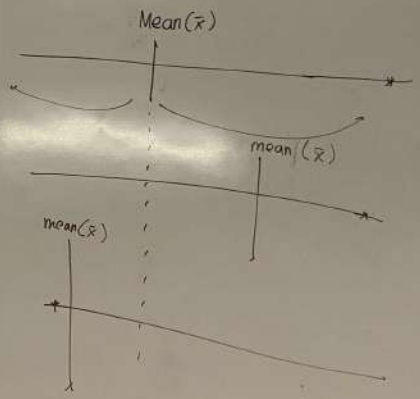
variabel / Nilai UTS	Body temperature	hair color
78	36,2°	black
79	38,1°	brown
100	35°	red
32	37°	black
10		grey

data


wed right
ar color
Variabel
Qualitative → non numerik
ID, gender, major
Quantitative → numerik
GPA, semester, & units
2 3 4 5 6 7 1000
$\bar{X} = \frac{2+3+4+5+6}{5} = 4$
$\bar{X} = 5,4$
Median = 4
Median = 4,5
Stabil = ROBUST.
Median
The median of a set of measurements is the middle measurement when the measurements are ranked from smallest to largest.
The position of the median is
.5(n + 1)
once the measurements have been ordered.
Organizing Variab
continuous data
17,23
17,2

VOC


skewed right
mean > median
skewed left

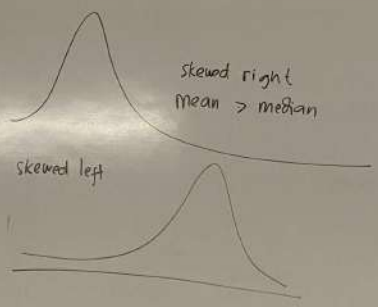

Mean($\bar{x}$)
mean($\bar{x}$)
mean($\bar{x}$)

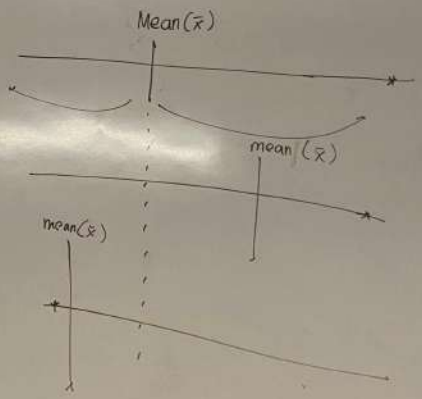
VOC

10

range
max - min
100 - 10 = 90

100





discrete = bil. bulat
integer

skewed right

skewed left

alat ukur

Variabel
Nilai UTS
78
79
100
32
10
} data

Body temperature
36,2°
38,1°
35°
37°

hair color
black
brown
red
black
grey



Variabel
Qualitative → non numerik
ID, gender, major
Quantitative → numerik
GPA, semester, # units

non angka → Gender
Major

angka → ID, GPA, semester, # of units

Gender
2
2
1
1
2
angka

1 = M
2 = F

TIDAK UNTUK DI TULIS

BINUS UNIVERSITY

Shapes

Mound shaped and symmetric (mirror images)

Skewed right: a few unusually large measurements

Skewed left: a few unusually small measurements

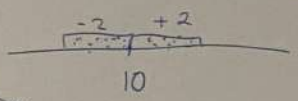
Bimodal: two local peaks

$S^2 = 4$

$S = 2$ vektor

arah simpangan

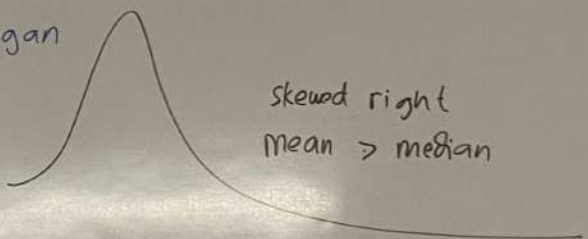
VOC



Standard deviation

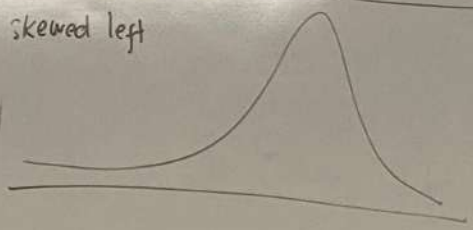
$S = \sqrt{S^2}$

$= (+)$



skewed right

mean > median



skewed left

$S^2 = 4$

$S = 2$ vektor

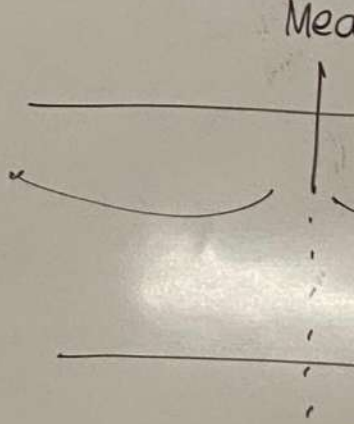
arah simpangan

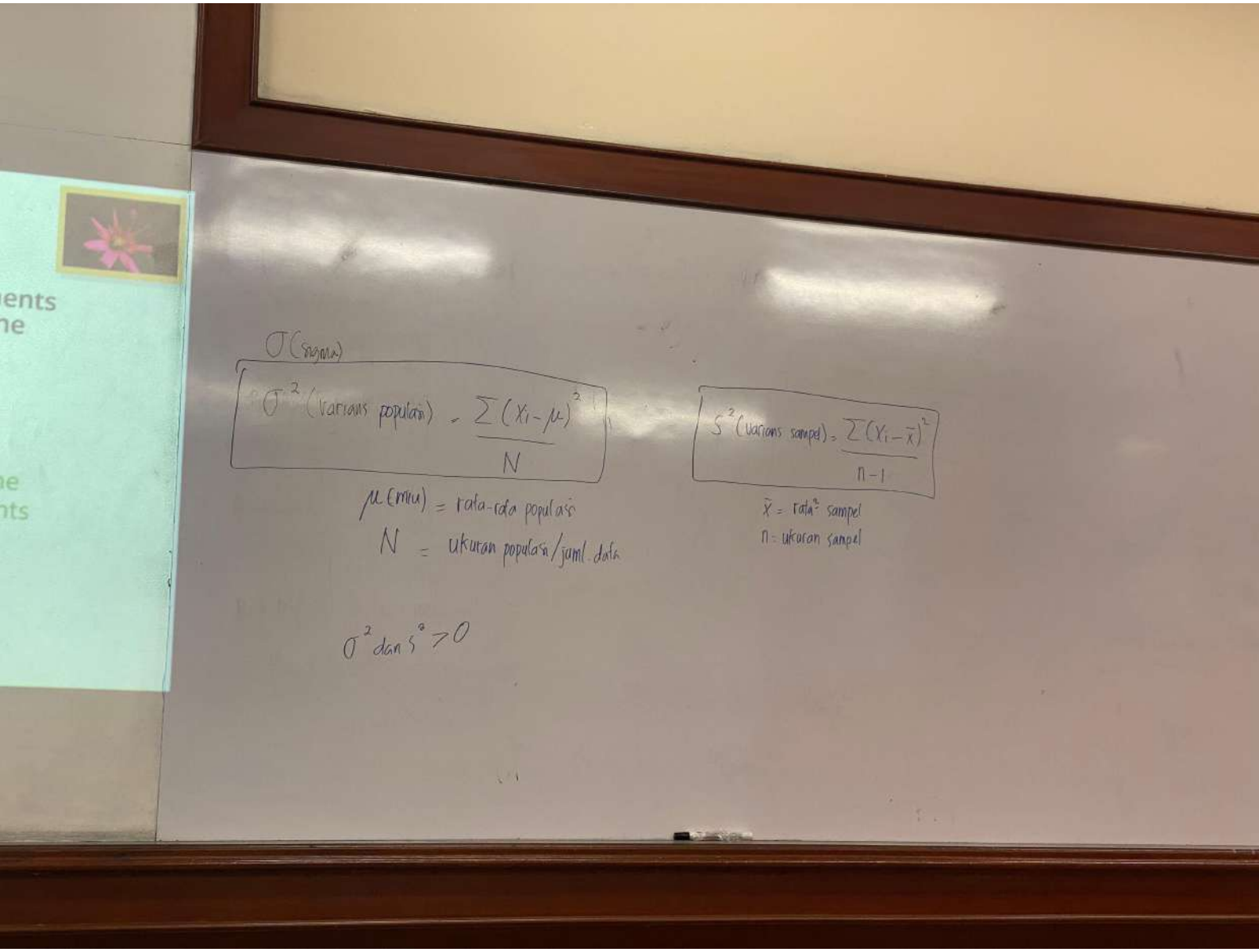
VOC

N = 100
Comp. eng
B29

n = 10

S^2
S




σ (sigma)
σ² (varians populasi) = Σ(Xi − μ)² / N
μ (mhu) = rata-rata populasi
N = ukuran populasi / jaml data
σ² dan s² > 0
s² (varians sampel) = Σ(Xi − x̄)² / n − 1
x̄ = rata² sampel
n = ukuran sampel

Basic Probability

- Peluang sebuah kejadian A, $P(A)$

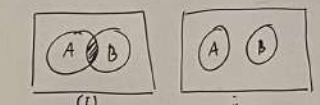
$$P(A) = \frac{n(A)}{n(S)}$$

$n(A)$ = banyaknya kejadian A

$n(S)$ = banyaknya sampel

$$0 \leq P(A) \leq 1$$

- Peluang 2 atau lebih kejadian



(I)

not mutually exclusive

(II)

mutually exclusive

- Dari gambar (I)

$A \cap B = \{ \ldots \}$

$P(A \cap B) = \ldots$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

- Dari gambar (II)

$A \cap B = \emptyset$

$P(A \cap B) = 0$

$P(A \cup B) = P(A) + P(B)$

JOINT and MARGINAL Probability

- digunakan untuk menentukan peluang dari 2 kejadian secara simultan.
- banyaknya kejadian ditampilkan dalam sebuah tabel kontingensi

Gender \ Status	Lulus (L)	Tidak lulus (TL)	
F	n_1	n_2	$n(F)$
M	n_3	n_4	$n(M)$
	$n(L)$	$n(TL)$	n_{Total}

JOINT PROBABILITY

$P(F \cap L) = \frac{n_1}{n_{Total}}$; $P(M \cap TL) = \frac{n_4}{n_{Total}}$

$P(M \cap L) = \frac{n_3}{n_{Total}}$

$P(F \cap TL) = \frac{n_2}{n_{Total}}$

MARGINAL PROB

$P(L) = \frac{n(L)}{n_{Total}}$ $P(F) = \frac{n(F)}{n_{Total}}$

$P(TL) = \frac{n(TL)}{n_{Total}}$ $P(M) = \frac{n(M)}{n_{Total}}$

Conditional Probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(B|A) = \frac{P(B \cap A)}{P(A)}$$

$P(A|B) \neq P(B|A)$

$P(A \cap B) = P(B \cap A)$

Independence

A dan B dikatakan independan (saling bebas) jika

1. $P(A|B) = P(A)$
2. $P(B|A) = P(B)$
3. $P(A \cap B) = P(A) \cdot P(B)$

Basic Probability

- Peluang sebuah kejadian A, $P(A)$

$$P(A) = \frac{n(A)}{n(S)}$$

$n(A)$ = banyaknya kejadian A

$n(S)$ = banyaknya sampel

$$0 \leq P(A) \leq 1$$

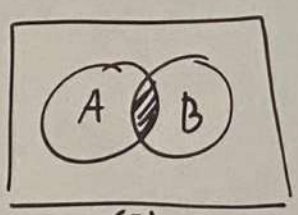
$n(S) = 100$ orang

A = mhs berkacamata

$n(A) = 25$

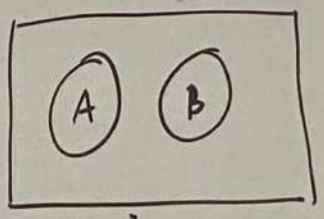
$$P(A) = \frac{n(A)}{n(S)} = \frac{25}{100} = 0{,}25 = \frac{1}{4}$$

- Peluang 2 atau lebih kejadian



(I)

not Mutually exclusive



II

Mutually exclusive

- Dari gambar (I)

$A \cap B = \{ \ldots \}$

$P(A \cap B) = \ldots$

$P(A \cup B) = P(A) +$

- Dari gambar (II)

$A \cap B = \emptyset$

$P(A \cap B) = 0$

$P(A \cup B) = P(A) + P$

Basic Probability

- Peluang sebuah kejadian A, $P(A)$

$$P(A) = \frac{n(A)}{n(S)}$$

$n(A)$ = banyaknya kejadian A ; $n(S)$ = banyaknya sampel

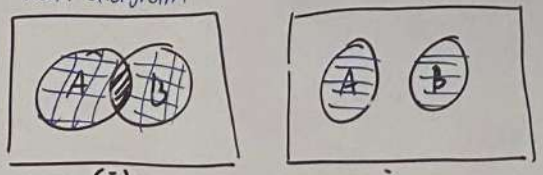
$$0 \leq P(A) \leq 1$$

$n(S)$ = 100 orang
A = mhs berkacamata
$n(A)$ = 25

$$P(A) = \frac{n(A)}{n(S)} = \frac{25}{100} = 0{,}25 = \frac{1}{4}$$

- Peluang 2 atau lebih kejadian

Venn diagram



(I) (II)

not Mutually exclusive
has intersection

Mutually exclusive
Saling lepas

- Dari gambar (I)

$A \cap B = \{\ldots\}$

$P(A \cap B) \neq 0$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

- Dari gambar (II)

$A \cap B = \emptyset$ (himpunan kosong)

$P(A \cap B) = 0$

$P(A \cup B) = P(A) + P(B)$

JOINT a
- digunak
- banyakn

Gende
F
M

JOIN
P(F
P(M∩
P(F∩

Basic Probability

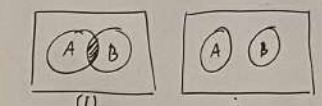
- Peluang sebuah kejadian A, $P(A)$

$$P(A) = \frac{n(A)}{n(S)}$$

$n(A)$ = banyaknya kejadian A
$n(S)$ = banyaknya sampel

$$0 \leq P(A) \leq 1$$

- Peluang 2 atau lebih kejadian



(i) not mutually exclusive

(ii) mutually exclusive

- Dari gambar (i)

$A \cap B = \{ \ldots \}$

$P(A \cap B) = \ldots$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

- Dari gambar (ii)

$A \cap B = \emptyset$

$P(A \cap B) = 0$

$P(A \cup B) = P(A) + P(B)$

JOINT and MARGINAL Probability

- digunakan untuk menentukan peluang dari 2 kejadian secara simultan.
- banyaknya kejadian ditampilkan dalam sebuah tabel kontingensi

	Status		
Gender	Lulus (L)	Tidak lulus (TL)	
F	n_1	n_2	$n(F)$
M	n_3	n_4	$n(M)$
	$n(L)$	$n(TL)$	n_{Total}

JOINT PROBABILITY

$$P(F \cap L) = \frac{n_1}{n_{Total}} \quad ; P(M \cap TL) = \frac{n_4}{n_{Total}}$$

$$P(M \cap L) = \frac{n_3}{n_{Total}}$$

$$P(F \cap TL) = \frac{n_2}{n_{Total}}$$

MARGINAL PROB

$$P(L) = \frac{n(L)}{n_{Total}} \qquad P(F) = \frac{n(F)}{n_{Total}}$$

$$P(TL) = \frac{n(TL)}{n_{Total}} \qquad P(M) = \frac{n(M)}{n_{Total}}$$

Conditional Probability (peluang bersyarat)

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(B|A) = \frac{P(B \cap A)}{P(A)}$$

$P(A|B) \neq P(B|A)$?

$P(A \cap B) = P(B \cap A)$

$P(A|B)$ = mencari peluang A jika diketahui B

$P(B|A)$ = mencari peluang B jika diketahui A.

Independence

A dan B dikatakan independen (saling bebas) jika

1. $P(A|B) = P(A)$ ✓
2. $P(B|A) = P(B)$ ✓
3. $P(A \cap B) = P(A) \cdot P(B)$ ✓

A dan B independen

$P(\ldots|\ldots)$

↓ dicari — ditanyakan

↓ diketahui — syarat

$P(\text{Lulus}|L)$

binusianorg.my.sharepoint.com/personal/rinda_hariswan_binus_ac_id/Documents/e%20book/%5BDavid_R_Anderson_Dennis_J_Sweeney_Th...

Activity	Participants (millions) Male	Female
Bicycle riding	22.2	21.0
Camping	25.6	24.3
Exercise walking	28.7	57.7
Exercising with equipment	20.4	24.4
Swimming	26.4	34.4

a. For a randomly selected female, estimate the probability of participation in each of the sports activities.
b. For a randomly selected male, estimate the probability of participation in each of the sports activities.
c. For a randomly selected person, what is the probability the person participates in exercise walking?
d. Suppose you just happen to see an exercise walker going by. What is the probability the walker is a woman? What is the probability the walker is a man?

C. Calvin

	Male	Female	
Bicyle	22,2	21,0	43,2
Camping	25,6	24,3	49,9
Walking	28,7	57,7	86,4
Equipment	20,4	24,4	44,8
Swiming	26,4	34,4	60,8
	123,3	161,8	285,1

$$P(\text{walking}) = \frac{86,4}{285,1} = 0,303$$ //

Basic Probability

- Peluang sebuah kejadian A, $P(A)$

$$P(A) = \frac{n(A)}{n(S)}$$

$n(A)$ = banyaknya kejadian A

$n(S)$ = banyaknya sampel

$$0 \leq P(A) \leq 1$$

$n(S) = 100$ orang

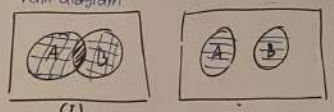
A = mhs berkacamata

$n(A) = 25$

$P(A) = \frac{n(A)}{n(S)} = \frac{25}{100} = 0{,}25 = \frac{1}{4}$

- Peluang 2 atau lebih kejadian

Venn diagram



(I)

not mutually exclusive
Ada irisan/intersection

II

Mutually exclusive
Saling lepas

- Dari gambar (I)

$A \cap B = \{\ldots\}$

$P(A \cap B) \neq 0$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

- Dari gambar (II)

$A \cap B = \emptyset$ (himpunan kosong)

$P(A \cap B) = 0$

$P(A \cup B) = P(A) + P(B)$

JOINT and MARGINAL Probability

- digunakan untuk menentukan peluang dari 2 kejadian secara simultan.
- banyaknya kejadian ditampilkan dalam sebuah tabel kontingensi

Gender \ Status	Lulus (L)	Tidak lulus (TL)	
F	n_1	n_2	$n(F)$
M	n_3	n_4	$n(M)$
	$n(L)$	$n(TL)$	n_{Total}

Frekuensi

JOINT PROBABILITY

$P(F \cap L) = \frac{n_1}{n_{Total}}$; $P(M \cap TL) = \frac{n_4}{n_{Total}}$

$P(M \cap L) = \frac{n_3}{n_{Total}}$

$P(F \cap TL) = \frac{n_2}{n_{Total}}$

MARGINAL PROB

$P(L) = \frac{n(L)}{n_{Total}}$ $P(F) = \frac{n(F)}{n_{Total}}$

$P(TL) = \frac{n(TL)}{n_{Total}}$ $P(M) = \frac{n(M)}{n_{Total}}$

a. Joshua

Total Female = 161.8

$P(\text{Bicycle ride}) = \frac{21}{161.8} = 0.129$

$P(\text{Camping}) = \frac{24.3}{161.8} = 0.15$

$P(\text{Walking}) = \frac{57.7}{161.8} = 0.356$

$P(\text{Exercise with equipment}) = \frac{24.4}{161.8} = 0.151$

$P(\text{Swimming}) = \frac{34.4}{161.8} = 0.213$

Yoga

b. Total Male = 123,3

P(Bicycle ride) = $\frac{22,2}{123,3} = 0,180$

P(camping) = $\frac{25,6}{123,3} = 0,207$

P(Walking) = $\frac{28,7}{123,3} = 0,232$

P(Exercise with Equipment) = $\frac{26,4}{123,3} = 0,165$

P(Swimming) = $\frac{26,4}{123,3} = 0,214$

29) Nickson 2402579751

	Yes	NO	
18 to 34	750	170	920 n(18 to 34)
35 & older	950	130	1080 n(35 ∧ older)
	1700 n(Yes)	300 n(No)	2000 n(total)

$P(18 \text{ to } 34 \cap Yes) = \frac{750}{2000} = 0.375$

$P(18 \text{ to } 34 \cap NO) = 170/2000 = 0.085$

$P(35 \text{ \& older} \cap Yes) = \frac{950}{2000}\ 0.475$

$P(35 \text{ \& older} \cap NO) = \frac{130}{2000} = 0.065$

d. kynan

$P(\text{woman} \cap \text{walking}) = \frac{\text{Sample}}{\text{total}} \rightarrow$ woman exercise walker: $\frac{57.7}{86.4} = 0.667 \approx 2/3$ chance.

male exercise walker: :D

$P(\text{man} \cap \text{walking}) = \frac{28.7}{86.4} = 0.333 \approx 1/3$ chance.

$n(\text{walking}) = 28.7 + 57.7 = 86.4$

E.) Facts

35< no health insurance

P.) $\left(\begin{matrix}\text{no health} \\ \text{insurance}\end{matrix} \middle| \begin{matrix}35 \\ \text{above}\end{matrix}\right) = \frac{130}{950 + 130}$

$= 0,12$

c

d) P(No asuransi)
18-34

total Yes = 750
total No = 170
> + = 920

$$P(\text{No} / 18\text{-}34) = \frac{170}{920}$$

$$= \frac{17}{92}$$

$$\approx 0.184$$

F) $P(\text{no insurance}) = \frac{300}{2000}$

170
130
300

$= 0,15$

$P(18-34) = \frac{170+750}{2000}$

$= 0,46$

$P(\ldots \mid \ldots)$

↓ ditanya ↓ diket

malvin

$P(\text{18-34} \mid \text{no ins}) = \frac{P(18-34 \cap \text{no})}{P(\text{No})}$

$= \frac{170}{300} = 0,57$

d) $P(\text{No asuransi}$

18-34

total Yes = 75

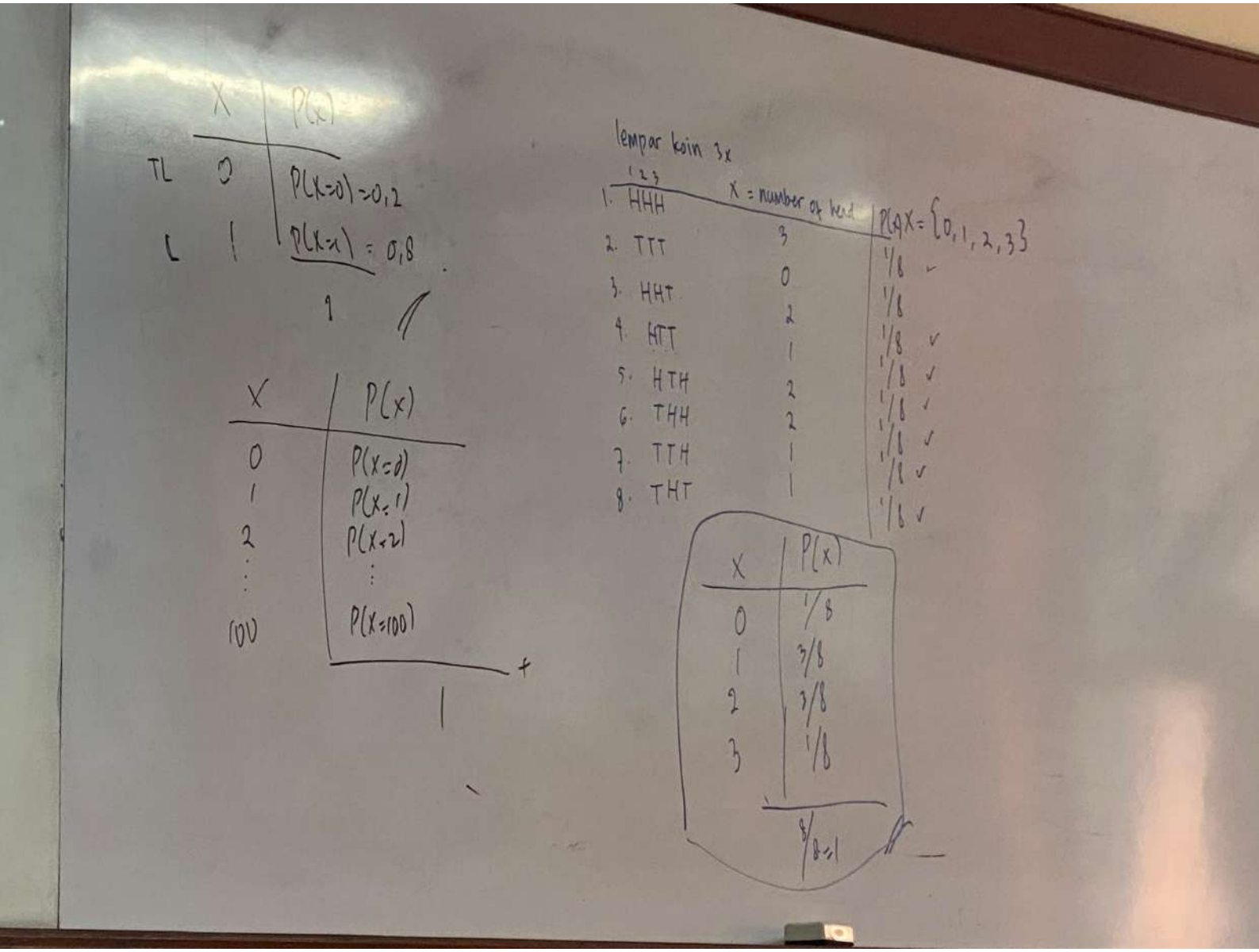
total No = 17

$P(\text{No} / 18-3$

C. Total yg tidak health insurance: 170 + 130
= 300

Total yes dan no = 300 + 1700
= 2000

$P(No) = \frac{300}{2000} = 0{,}15 \rightarrow 15\%$


X | P(x)
TL 0 | P(X=0) = 0,2
L 1 | P(X=1) = 0,8
1
X | P(x)
0 | P(X=0)
1 | P(X=1)
2 | P(X=2)
100 | P(X=100)
1
lempar koin 3x
1 2 3
1. HHH
2. TTT
3. HHT
4. HTT
5. HTH
6. THH
7. TTH
8. THT
X = number of head
3
0
2
1
2
2
1
1
P(X) = {0, 1, 2, 3}
1/8 ✓
1/8
1/8 ✓
1/8 ✓
1/8 ✓
1/8 ✓
1/8 ✓
1/8 ✓
X | P(x)
0 | 1/8
1 | 3/8
2 | 3/8
3 | 1/8
8/8 = 1

Random variable → fungsi yg memetakan outcome sebuah percobaan
- himpunan
- dinyatakan dgn huruf kapital
- dalam bentuk kuantitatif

X = status akhir mahasiswa dlm matkul probstat

X = {lulus, ngulang (TL)}

$X = \{1, 0\}$

X = jumlah HP yg dimiliki oleh mhs

$X = \{0, 1, 2, 3, \dots\}$ → discrete number

$\Sigma \rightarrow x=0+x=1+x=\dots$ $P(X \leq 2) = P(X=2)+P(X=1)+P(X=0)$

$\sum_{i=0}^{2}$

Z = nilai UTS mahasiswa

$Z = \{0; 10,2; 75; 100\}$ → continous

$0 \leq Z \leq 100$ ← interval range.

integral.

$P(X \leq 2) = \int_{0}^{2} f(x)\,dx$

P(A) X = {0, 1, 2, 3}

1/8 ✓
1/8
1/8
1/8 ✓
1/8 ✓
1/8 ✓
1/8 ✓
1/8 ✓
1/8 ✓
1/8 ✓

- cdf = cummulative density function
  $P(X \leq x) = F(x)$

- pmf = probability mass function
  $f(x)$ → fs. peluang

Random variable → fungsi yg memetakan outcome sebuah percobaan

- himpunan
- dinyatakan dgn huruf kapital
- dalam bentuk kuantitatif

X = status akhir mahasiswa dlm matkul probstat

X = {lulus, ngulang (TL)}

$X = \{1, 0\}$

X = jumlah HP yg dimiliki oleh mhs

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$Z = \{0; 10,2; 75; 100\}$ → continous

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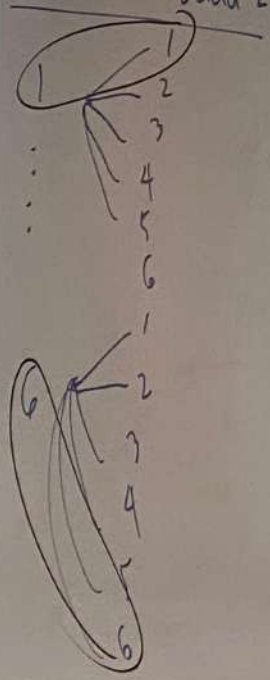
	X	P(x)
TL	0	P(X=0) = 0,2
L	1	P(X=1) = 0,8
		1

lempar koin 3x

	1 2 3	X = number of head	P(x)
1.	HHH	3	1/8 ✓
2.			1/8
3.			1/8 ✓

$P(x) = \{0, 1, 2, 3\}$

Dadu 1 Dadu 2



$X = 0$ (1,1); (2,2); (3,3) (4,4); (5,5); (6,6)

$X = 1$ (1,2); (2,1); (2,3) (3,2); (3,4); (4,3) (4,5); (5,4); (5,6) (6,5)

$X = 2 \rightarrow$ (1,3); (3,1); (2,4); (4,2) (5,3); (3,5); (4,6); (6,4)

$X = 3 \rightarrow$ (1,4); (4,1); (2,5); (5,2) (3,6); (6,3)

$X = 4$, (1,5), (5,1), (4,2); (2,6)

$X = 5 \rightarrow$ (1,6); (6,1)

- cdf = cummulative density
  $P(X \leq x) = F(x)$
- pmf = probability mass funct
  $f(x) \rightarrow$ fs. peluang

(3) Discrete Random Variable (Calvin Prince)

- Nilai varians dari variabel random diskrit

$$Var(x) = \sigma^2 = E(x^2) - [E(x)]^2$$

$$= \sum_{i=1}^{n} x_i^2 \cdot P(x_i) - [E(x)]^2$$

contoh soal:

X_i = selisih positif dari 2 mata dadu

(1)

X	P(Xi)	x^2
0	6/36	0
1	10/36	1
2	8/36	4
3	6/36	9
4	4/36	16
5	2/36	25
	36/36	//

1. Lengkapi tabel distribusi peluang
2. Hitung nilai rata-rata (mean)
3. Hitung nilai varians (σ^2)

(Kiyan Gn)

$(2902584453)(2)$ / $= \sum X_i \cdot P(x_i)$

$= 0 \cdot 6/36 = 0$

$1 \cdot 10/36 = 10/36$

$2 \cdot 8/36 = 16/36$

$3 \cdot 6/36 = 18/36$

$4 \cdot 4/36 = 16/36$

$5 \cdot 2/36 = 10/36$ +

$70/36$

$\mu = 1.945.$

$\sum X_i^2 \cdot P(x_i)$

$= 0 \cdot 6/36 = 0$

$1 \cdot 10/36 = 10/36$

$4 \cdot 8/36 = 32/36$

$9 \cdot 6/36 = 54/36$

$16 \cdot 4/36 = 64/36$

$25 \cdot 2/36 = 50/36$ +

$220/36$

$5,833$

$[E(x)] = 1,945$

$[E(X)]^2 = 3,787$

$\sigma^2 = \sum X_i^2 \cdot P(x_i) - [E(x)]^2 = 5,833 - 3,787$

$= 2,05//$

DISTRIBUSI BINOMIAL

- Merupakan fungsi distribusi untuk var. random diskrit
- Distribusi binomial memiliki karakteristik :
  1. berlaku pada percobaan yang hanya memiliki 2 outcome
  2. Melibatkan sampel berukuran n
  3. antar sampel / pengamatan independen
  4. peluang salah satu outcomenya PASTI DIKETAHUI
- Misal :

  X = var. random berdistribusi Binomial dengan sampel sebanyak n dan peluang salah satu outcomenya diketahui, Maka dapat ditulis :

$$X \sim B(n, p)$$

parameter

- Fungsi peluang binomial

$$P(X = x) = f(x) = \binom{n}{x} \cdot p^x \cdot (1-p)^{n-x}$$

- Nilai rata-rata peluang Binomial

$$\mu = E(x) = n.p$$

- Nilai variansi

$$\sigma^2 = n.p.(1-p)$$

contoh :

Diketahui pada sebuah ujian kelas statistik, peluang mahasiswa lulus adalah 0,4. Dari 8 mahasiswa, hitunglah :

1. peluang 3 mahasiswa lulus
2. peluang ada 2-4 mahasiswa lulus
3. peluang paling banyak 3 mahasiswa lulus
4. peluang paling sedikit 8 mahasiswa lulus.

Discrete Random Variable

- $X = \{0, 1, 2, \dots\}$

  dimana nilai anggota himpunan X adalah Bilangan Bulat

- $f(x)$ = fungsi peluang (pmf)

  $F(x) = P(X \leq x)$ = fungsi kumulatif (cdf)

- Nilai rata² atau mean dari variabel random diskrit.

  $E(x)$ = expected value = μ

  $$E(x) = \mu = \sum_{i=1}^{n} X_i \cdot P(X_i)$$

- Nilai varians dari variabel random diskrit

  $$\text{Var}(x) = \sigma^2 = E(x^2) - [E(x)]^2$$

  $$= \sum_{i=1}^{n} x_i^2 \cdot P(x_i) - [E(x)]^2$$

contoh soal:

X = banyaknya mahasiswa lulus ; n = 8

X = {0, 1, 2, 3, ..., 8}

1. $P(X=3) = \ldots$

2. $P(2 \le x \le 4) = P(X=2) + P(X=3) + P(X=4)$
$= 0,209 + 0,278 + 0,232 = \ldots$
$= 0,72$ Hendra (2902804091)

3.) $x = \{0, 1, 2, 3, 4, 5, 6, 7, 8\}$

$P(x=0,1,2,3)$

$f(3) = \binom{8}{3} \cdot 0,4^3 \cdot 0,6^5$
$= 56 \cdot 0,4^3 \cdot 0,6^5$
$= 0,278$

$f(2) = \binom{8}{2} \cdot 0,4^2 \cdot 0,6^6$
$= 28 \cdot 0,4^2 \cdot 0,6^6$
$= 0,209$

$f(1) = \binom{8}{1} \cdot 0,4^1 \cdot 0,6^7$
$= 8 \cdot 0,4^1 \cdot 0,6^7$
$= 0,089$

$f(0) = \binom{8}{0} \cdot 0,4^0 \cdot 0,6^8$
$= 1 \cdot 0,4^0 \cdot 0,6^8$
$= 0,016$

$\therefore P(x \le 3) = 0,592$

faris rafa firdaus
(2902614294)

$P(X=2) = f(2) = \binom{8}{2} 0,4^2 (1-0,4)^{8-2}$
$= 28 \cdot 0,16 \cdot 0,046656$
$= 0,209$

$P(X=4) = f(4) = \binom{8}{4} 0,4^4 (1-0,4)^{8-4}$
$= 70 \cdot 0,0256 \cdot 0,1296$
$= 0,232$

4. $P(X=8) = f(8) = \binom{8}{8} \cdot 0,4^8 \cdot 0,6^0$
$= 1 \cdot 0,00066 \cdot 1$
$= 0,00066$

$\frac{n!}{x!(n-x)!} = \frac{8!}{2!(8-2)!} \quad \frac{8!}{4!4!} \rightarrow \frac{8 \cdot 7 \cdot 6 \cdot 5}{4 \cdot 3 \cdot 2 \cdot 1} = 70$

$= \frac{8!}{2! \cdot 6!}$

$= \frac{8 \cdot 7 \cdot 6!}{2 \cdot 1 \cdot 6!}$

$= \frac{56}{2} = 28$

b) At least four:

$P(X \geq 4) = 1 - (P(X=0) + P(X=1) + P(X=2) + P(X=3))$

$P(X=0) = \binom{10}{0} 0.25^{0} (1-0.25)^{10-0} = 0.056$

$P(X=1) = \binom{10}{1} 0.25^{1} (1-0.25)^{10-1} = 0.188$

$P(X=2) = \binom{10}{2} 0.25^{2} (1-0.25)^{10-2} = 0.282$

$P(X=3) = \binom{10}{3} 0.25^{3} (1-0.25)^{10-3} \sim 0.250$

0.776

$P(X \geq 4) = 1 - 0.776 = \underline{0.224}$

a) $X \sim B(10, 0.25)$

$n = 10$

$p = 0.25$

$q = 1 - 0.25 = 0.75$

$P(X=5) = \binom{10}{5} \cdot (0.25)^{5} \cdot (0.75)^{5}$

$= 252 \times 0.000998 \times 0.237$

$= 0.0584$

d) $X \sim B(10, 025)$

$P(X \leq 2) = P(x=0) + P(x=1) + P(x=2)$

$P(x=0) = \binom{10}{0} (0.25)^{0} (0.75)^{10} = 0.056$

$P(x=1) = \binom{10}{1} \cdot (0.25)^{1} \cdot (0.75)^{9} = 0.188$

$P(x=2) = \binom{10}{2} \cdot (0.25)^{2} \cdot (0.75)^{8} = 0.282$

$\rightarrow P(x \leq 2) = 0.056 + 0.188 + 0.282$

$= 0.526$

Wulandari Medina R - 2902596960

c) $X \sim B(20, 0.25)$

$n = 20$ (Calvin Prince (29026338))

$p = 0,25$

$q = 1 - 0,25 = 0,75$

$P(x=0) = \binom{20}{0} \cdot (0,25)^{0} \cdot (0,75)^{20}$

$= 1 \times 1 \times 0,056 = 0,056$ //

$X = \{0, 1, 2, 3, 4, 5, 6, 7, 8\}$

$X \sim B(8, 0.4)$

1. $P(X=3) = f(3) = \binom{8}{3} \cdot 0.4^{3} \cdot 0.6^{5}$

$= 56 \cdot 0.64 \cdot 0.077$

$= 0.279$

Joshua - 2902606670

DISTRIBUSI POISSON

- digunakan untuk variabel random yg menyatakan jumlah / banyaknya
- X = banyaknya kendaraan yg melewati GT Halim
  Y = jumlah gangguan koneksi internet
- $X \sim Poisson(\lambda)$ dimana λ = rata-rata kejadian (parameter)
- Fungsi peluang Poisson:

$$P(X=x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!}$$

Contoh soal

Pada sebuah proses penerbitan sebuah buku bacaan diketahui pada 1 halaman rata-rata terdapat 3 kesalahan ketik (typo). Hitunglah peluang jika terdapat:

1. 5 kesalahan ketik
2. paling banyak 3 kesalahan ketik
3. paling sedikit 2 kesalahan ketik.

X = banyaknya kesalahan ketik

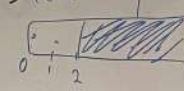
$X = \{0, 1, 2, 3, 4, \ldots\}$

$X \sim Poisson(3)$

1. $P(X=5) = \frac{e^{-\lambda} \cdot \lambda^x}{x!}$

$= \frac{e^{-3} \cdot 3^5}{5!} = 0{,}202$

3. $P(X \geq 2) = P(X=2) + P(X=3) + \ldots$



$P(X \geq 2) = 1 - P(X < 2)$

$= 1 - [P(X=1) + P(X=0)] = 1 - (0{,}149 + 0{,}049)$

$= 1 - 0{,}198 = 0{,}802$

$P(X=0) = \frac{e^{-3} \cdot 3^0}{0!} = 0{,}049$

$P(X=1) = \frac{e^{-3} \cdot 3^1}{1!} = 0{,}149$

2. $P(X \leq 3)$, $\lambda = 3$ ke-sha (nim Versen 2902737590)

$P(X=3) + P(X=2) + P(X=1) + P(X=0)$

- $P(X=3) = \frac{e^{-3} \cdot 3^3}{3!} = 0{,}224$
- $P(X=2) = \frac{e^{-3} \cdot 3^2}{2!} = 0{,}224$
- $P(X=1) = \frac{e^{-3} \cdot 3^1}{1!} = 0.149$
- $P(X=0) = \frac{e^{-3} \cdot 3^0}{0!} = 0.049$

$0{,}224 + 0{,}224 + 0.149 + 0.0498 = 0.6468$

Aisya (2902606960)

b) At least four:

$P(X \geq 4) = 1 - (P(X=0) + P(X=1) + P(X=2) + P(X=3))$

$P(X=0) = \binom{10}{0} 0.25^{0} (1-0.25)^{10-0} = 0.056$

$P(X=1) = \binom{10}{1} 0.25^{1} (1-0.25)^{10-1} = 0.188$

$P(X=2) = \binom{10}{2} 0.25^{2} (1-0.25)^{10-2} = 0.282$

$P(X=3) = \binom{10}{3} 0.25^{3} (1-0.25)^{10-3} = 0.250$

0.776

$P(X \geq 4) = 1 - 0.776 = \underline{0.224}$

a) $X \sim B(10, 0.25)$

$n = 10$

$p = 0.25$

$q = 1 - 0.25 = 0.75$

$P(X=5) = \binom{10}{5} \cdot (0.25)^{5} \cdot (0.75)^{5}$

$= 252 \times 0.000098 \times 0.237$

$= 0.0584$

Contoh:

Diketahui pada sebuah ujian kelas statistika, peluang mahasiswa lulus adalah (0.4) Dari (8) mahasiswa, hitunglah:

1. peluang 3 mahasiswa lulus
2. peluang ada 2-4 mahasiswa lulus
3. peluang paling banyak 3 mahasiswa lulus
4. peluang paling sedikit 8 mahasiswa lulus.

$X = \{0, 1, 2, 3, 4, 5, 6, 7, 8\}$

$X \sim B(8, 0.4)$

1. $P(X=3) = f(3) = \binom{8}{3} \cdot 0.4^{3} \cdot 0.6^{5}$

$= 56 \cdot 0.64 \cdot 0.077$

$= 0.279$

Jochua - 2902606670

Contoh soal!

1. Buktikan bahwa fungsi di bawah ini merupakan sebuah pdf!

$$f(x) = \begin{cases} 0{,}15\, e^{-0{,}15(x-0{,}5)} & ; x \geq 0{,}5 \\ 0 & ; x \text{ lainnya} \end{cases}$$

- Mean dan varians variabel random kontinyu.

⇒ mean (μ)

$$\mu = E(x) = \int_{-\infty}^{\infty} x \cdot f(x)\, dx$$

⇒ varians (σ^2)

$$\sigma^2 = \text{var}(x) = \int_{-\infty}^{\infty} (x-\mu)^2 \cdot f(x)$$

Contoh soal!

1. Buktikan bahwa fungsi di bawah ini merupakan sebuah pdf!

$$f(x) = \begin{cases} 0{,}15\, e^{-0{,}15(x-0{,}5)} & ; x \geq 0{,}5 \\ 0 & ; x \text{ lainnya} \end{cases}$$

- Mean dan varians variabel random kontinyu.

→ mean (μ)

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→ varians (σ^2)

$$\sigma^2 = \text{var}(x) = \int_{-\infty}^{\infty} (x-\mu)^2 \cdot f(x)\, dx$$

$$\sigma^2 = \text{var}(x) = E(x^2) - [E(x)]^2$$

$$= \int_{-\infty}^{\infty} x^2 \cdot f(x)\, dx - [E(x)]^2$$

Contoh soal:

1. Buktikan bahwa fungsi di bawah ini merupakan sebuah pdf!

$$f(x) = \begin{cases} 0{,}15\, e^{-0{,}15(x-0{,}5)} & ; x \geq 0{,}5 \\ 0 & ; x \text{ lainnya} \end{cases}$$

- Mean dan varians variabel random kontinu

→ mean (μ)

$$\mu = E(x) = \int_{-\infty}^{\infty} x \cdot f(x)\, dx$$

→ varians (σ^2)

$$\sigma^2 = \text{var}(x) = \int_{-\infty}^{\infty} (x-\mu)^2 \cdot f(x)\, dx$$

$$\sigma^2 = \text{var}(x) = E(x^2) - [E(x)]^2$$

$$= \int_{-\infty}^{\infty} x^2 \cdot f(x)\, dx - [E(x)]^2$$

Contoh soal:

2. Diketahui fungsi peluang sebagai berikut:

$$f(x) = \begin{cases} 0{,}75x + 0{,}2 & ; 3 \leq x \leq 5 \\ 0 & ; x \text{ lainnya} \end{cases}$$

a. Hitunglah $P(x \leq 4)$

b. Hitunglah $P(x > 4)$

c. $P(3{,}5 \leq x \leq 4{,}5)$

d. $P(x > 4{,}5)$

e. Mean (μ)

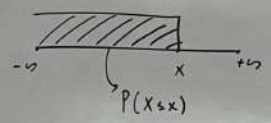
f. Varians (σ^2)

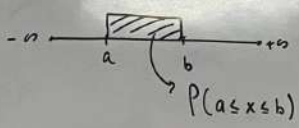
Continuous Random Variable

- Merupakan variabel random yg berbentuk bilangan real, $x \in R$.
- biasanya dinyatakan dalam bentuk interval / range.
- contoh:
  - X : nilai uts mahasiswa , $0 \leq x \leq 100$
  - Y : tinggi badan mahasiswa , $140 \leq y \leq 200$
  - Z : suhu ruangan di kampus syahdan, $20 \leq z \leq 38$

- Fungsi peluang var. random kontinyu

$P(X \leq x) = f(x) =$ pdf

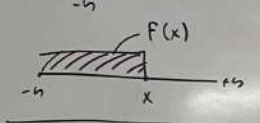


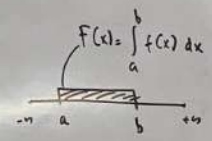


- Fungsi kumulatif var. random kontinyu

$F(x)$ = fungsi kumulatif (CDF)

$$F(x) = \int_{-\infty}^{x} f(x)\,dx$$





$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

Contoh soal:

1. Buktikan bahwa fungsi di bawah ini merupakan sebuah pdf!

$$f(x) = \begin{cases} 0{,}15\, e^{-0{,}15(x-0{,}5)} & , x \geq 0{,}5 \\ 0 & , x \text{ lainnya} \end{cases}$$

$$\int_{4{,}5}^{5} f(x)\,dx + \int_{5}^{\infty} 0\,dx$$

faris (2502614734)

2.) a.) $P(x \leq 4)$

$$\int_{3}^{4} 0{,}75x + 0{,}2\,dx$$

$$\left[\frac{0{,}75x^2}{2} + 0{,}2x\right]_3^4$$

$$= \left[\frac{0{,}75\,(4)^2}{2} + 0{,}2 \cdot 4\right] - \left[\frac{0{,}75\,(3)^2}{2} + 0{,}2 \cdot 3\right]$$

$$= 6{,}8 - 3{,}975$$

$$= 2{,}825$$

$$\int_{3}^{5} 0{,}75x + 0{,}2\,dx$$

$$\left[\frac{0{,}75x^2}{2} + 0{,}2x\right]_3^5$$

$$= \left[\frac{0{,}75\,(5)^2}{2} + 0{,}2\,(5)\right] - \left[\frac{0{,}75\,(3)^2}{2} + 0{,}2\,(3)\right]$$

$$= 10{,}375 - 3{,}975$$

$$= 6{,}4$$

$$\frac{2{,}825}{6{,}4} = 0{,}441$$

Contoh soal:

2. Diketahui fungsi pelwang sebagai berikut:

$$f(x) = \begin{cases} 0{,}75x + 0{,}2 & , 3 \leq x \leq 5 \\ 0 & , x \text{ lainnya} \end{cases}$$

a. Hitunglah $P(x \leq 4)$

b. Hitunglah $P(x > 4)$

c. $P(3{,}5 \leq x \leq 4{,}5)$

d. $P(x > 4{,}5)$

e. mean (μ)

f. variance (σ^2)

2b) $P(X>4)$

$\rightarrow \int_4^5 0.75x + 0.2 \, dx + \int_5^\infty 0 \, dx$

Five number summary:

1. Min
2. Q_1
3. Q_2 = Median
4. Q_3
5. Max

2 d) $P(X > 45)$

Calum Prince
(2 9026 33854

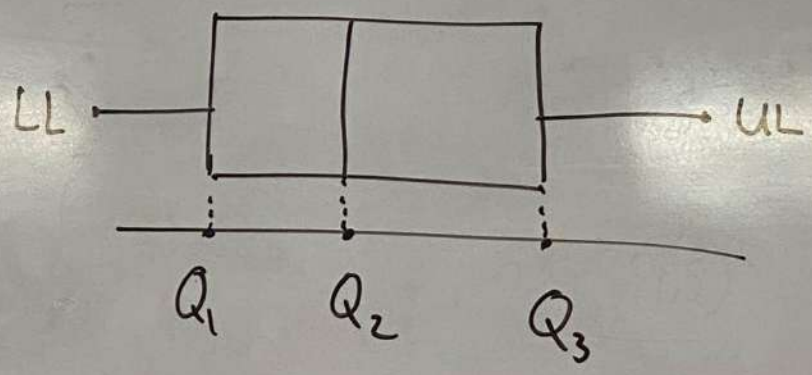
∞

$\int_0 dx$

5

$\int_0^{\infty} dx$

Boxplot



$LL = Q_1 - 1{,}5 \cdot IQR$

$UL = Q_3 + 1{,}5 \cdot IQR$

$IQR = Q_3 - Q_1$

2b) $P(X>4)$

$\int_4^5 0.75x + 0.2 \, dx + \int_5^\infty 0 \, dx$

$= \left[\frac{0.75x^2}{2} + 0.2x\right]_4^5 + 0$

$= \frac{0.75(25)}{2} + 0.2(5) + 0 - \frac{0.75(16)}{2} - 0.2(4) + 0$

$= 9.375 + 1 - 6 - 0.8 = 3.575$

$\int_3^5 0.75x + 0.2 \, dx = \left[\frac{0.75x^2}{2} + 0.2x\right]_3^5$

$= \frac{0.75(25)}{2} + 0.2(5) - \frac{0.75(9)}{2} - 0.2(3)$

$= 9.375 + 1 - 3.375 - 0.6 = 6.4$

$P(X>4) = 3.575/6.4 = 0.559$

2d) $P(X>4.5)$ (Calculus Primer)

$= \int_{4.5}^5 0.75x + 0.2 \, dx + \int_5^\infty 0 \, dx$

$= \left[\frac{0.75x^2}{2} + 0.2x\right]_{4.5}^5$

$= \left(\frac{0.75}{2}(5)^2 + 0.2(5)\right) - \left(\frac{0.75}{2}(4.5)^2 + 0.2(4.5)\right)$

$= 9.375 + 1 - 7.59375 - 0.9 = 1.88125$

$\int_3^5 0.75x + 0.2 \, dx = \left[\frac{0.75x^2}{2} + 0.2x\right]_3^5$

$= \left(\frac{0.75(5)^2}{2} + 0.2(5)\right) - \left(\frac{0.75(3)^2}{2} - 0.2(3)\right)$

$= 9.375 + 1 - 3.375 - 0.6 = 6.4$

$P(X>4.5) = \frac{1.88125}{6.4} = 0.294$

2c. $P(3.5 \leq X \leq 4.5)$

$= \int_{3.5}^{4.5} 0.75x + 0.2 \, dx + \int_{4.5}^\infty 0 \, dx$

$= \left[\frac{0.75x^2}{2} + 0.2x\right]_{3.5}^{4.5}$

$= \frac{0.75(4.5)^2}{2} + 0.2(4.5) + 0 - \left(\frac{0.75(3.5)^2}{2} + 0.2(3.5) + 0\right)$

$= 7.594 + 0.9 - 4.594 - 0.7 = 4.6$

$\int_3^5 0.75x + 0.2 \, dx = \left[\frac{0.75x^2}{2} + 0.2x\right]_3^5$

$= \left(\frac{0.75(25)}{2} + 0.2(5)\right) - \left(\frac{0.75(16)}{2} + 0.2(4)\right)$

$= 9.375 + 1 - 3.375 - 0.6 = 6.4$

$P(3.5 \leq X \leq 4.5) = \frac{4.6}{6.4} = 0.719$

contoh:

Diketahui kelebalan lapisan semikonduktor memiliki rata-rata 500 nm dan varians 64 nm. Seorang engineer mensyaratkan bahwa semikonduktor meme nuhi spesifikasi jika memiliki ketebalan antara 485 nm dan 515 nm.

X = ketebalan lapisan semikond

$X > 0$ maka

$X \sim N(500, 64)$ ✓

1. Tentukan peluang sebuah semikonduktor memenuhi spesifikasi.

2. Tentukan peluang sebuah semikonduktor memiliki ketebalan tidak lebih dari 510 nm.

$$\int \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

→ konstanta 18,...

1. $\mu = 500$ mm $\sigma = 8$ mm (Calvin Prince (29026331854)
$\sigma^2 = 64$ mm

$$P(X \leq x) = \int_{485}^{515} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx = \left(\frac{1}{\sqrt{2\pi \cdot 64}}\right)\left(\int_{485}^{515} e^{-\frac{1}{2}\left(\frac{x-500}{8}\right)^2} dx\right)$$

$= (0,049867778505)(28,833948122) = 0,939$ //

C. $X \sim N(15015, 12531600)$

$$P(X \leq x) = \int_{12000}^{18000} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

$$= \left(\frac{1}{\sqrt{2\pi\ 12531600}}\right)\left(\int_{12000}^{18000} e^{-\frac{1}{2}\left(\frac{x-15015}{3540}\right)^2} dx\right)$$

$= (1,127 \cdot 10^{-4})(5352.967)$

$= 0.6033$

D.) $X \sim N(15.015, 3540)$ faris (29026147294)

$$P(X \leq x) = \int_{0}^{14.000} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

$$= \frac{1}{\sqrt{2\pi\ 3540^2}} \int_{0}^{14.000} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

$$= \frac{1}{\sqrt{2\pi\ (12531.600)}} \int_{0}^{14.000} e^{-\frac{1}{2}\left(\frac{x-15.015}{3540}\right)^2} dx$$

$= (1,126955594 \cdot 10^{-4})(3435.370937)$

$= 0,387$

2. $X \sim N(500, 64)$

$$P(X \leq x) = \int_{0}^{510} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

$$= \frac{1}{\sqrt{2\pi \cdot 64}} \int_{0}^{510} e^{-\frac{1}{2}\cdot\left(\frac{x-500}{8}\right)^2}$$

$= (0.049861)(17.93492)$

$= 0.894$

$X \sim N(15.015, 12531600)$

b. $P(X < 10000) = \frac{1}{\sqrt{2\pi(12,531,600)}} \int_0^{10000} e^{-\frac{1}{2}\left(\frac{x-15,015}{3540}\right)^2} dx$

matlab

$69.6 \times \frac{1}{\sqrt{2 \cdot \pi \cdot (12,531,600)}}$

$= 0,078$

a) $P(X > 18000) = 1 - P(X \leq 18000)$

$= 1 - \int_0^{18000} \frac{1}{\sqrt{2\pi(12531600)}} e^{-\frac{1}{2}\left(\frac{x-15015}{3540}\right)^2} dx$

$X \sim N(15015, 12531600)$

$= 0,199564$

DISTRIBUSI NORMAL

- Merupakan salah satu distribusi peluang untuk variabel random kontinyu dengan domain $[-\infty, +\infty]$
- Contoh:
  X = nilai UTS Mahasiswa
  Maka $0 \leq x \leq 100$
  Y = berat bersih minuman kaleng (ml)
  Maka $0 \leq y \leq 600$

- Sebuah variabel random X berdistribusi Normal dengan rata-rata μ dan varians σ^2 dapat ditulis sebagai:

$$X \sim N(\underbrace{\mu, \sigma^2}_{\text{parameter}})$$

- Fungsi peluang dist. Normal

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

$$P(X \leq x) = \int_{-\infty}^{x} f(x)\,dx = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

$$\frac{1}{\sqrt{2\pi\sigma^2}} \int e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

→ konstanta 18,...

contoh:

Diketahui kelebalan lapisan semikonduktor memiliki rata-rata 500 nm dan varians 64 nm. Seorang engineer mensyaratkan bahwa semikonduktor memenuhi spesifikasi jika memiliki ketebalan antara 485 nm dan 515 nm.

1. Tentukan peluang sebuah semikonduktor memenuhi spesifikasi.
2. Tentukan peluang sebuah semikonduktor memiliki ketebalan tidak lebih dari 510 nm.

DISTRIBUSI NORMAL Baku / standar

- merupakan bentuk standaricasi dari distribusi Normal (μ, σ^2)
- variabel random $X \sim N(\mu, \sigma^2)$ diubah menjadi $Z \sim N(0,1)$ distandarkan melalui nilai $\mu = 0$ dan $\sigma^2 = 1$.
- Fungsi peluang dist. normal baku:

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(z)^2}$$

dimana $z = \frac{x-\mu}{\sigma}$

- Mencari nilai peluang dist. Normal melalui nilai standarisasi Z

1. ubah nilai x menjadi z

$$z = \frac{x-\mu}{\sigma}$$

2. Z hanya boleh 2 desimal

3. $P(X=x) = P(Z=z)$
   $P(X<x) = P(X \leq x) = P(Z<z) = P(Z \leq z)$] langsung lihat tabel Z
   $P(X>x) = P(Z>z) = 1 - \boxed{P(Z \leq z)}$
   $P(X \geq x) = P(Z \geq z) = 1 - \boxed{P(Z<z)}$ lihat nilai peluang di tabel Z

$= 1 - P(X \leq 18000)$ $\quad X \sim N(15015, 1254600)$

$= 1 - \int_{-\infty}^{18000} \frac{1}{} \; e^{-\frac{1}{2}\left(\frac{x-15015}{3540}\right)^2} dx$

DISTRIBUSI NORMAL Baku / standar

- merupakan bentuk standaricasi dari distribusi Normal (μ, σ)
- variabel random $X \sim N(\mu, \sigma^2)$ diubah menjadi $Z \sim N(0,1)$ distandarkan melalui nilai $\mu = 0$ dan $\sigma^2 = 1$.
- Fungsi peluang dist. normal baku:

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(z)^2}$$

dimana $z = \frac{x-\mu}{\sigma}$

- Mencari nilai peluang dist. Normal melalui nilai standarisasi Z

1. ubah nilai x menjadi z

$$z = \frac{x-\mu}{\sigma}$$

2. Z hanya boleh 2 desimal

3. $P(X=x) = P(Z=z)$
   - $P(X<x) = P(X \leq x) = P(Z<z) = P(Z \leq z)$] langsung lihat tabel Z
   - $P(X>x) = P(Z>z) = 1 - \boxed{P(Z \leq z)}$
   - $P(X \geq x) = P(Z \geq z) = 1 - \boxed{P(Z < z)}$] lihat nilai peluang di tabel Z

$= 1 - P(X \leq 18000)$

$X \sim N(15015, 1254600)$

$= 1 - \int_{-\infty}^{18000} \frac{1}{} e^{-\frac{1}{2}\left(\frac{x-15015}{1540}\right)^2} dx$

DISTRIBUSI NORMAL Baku / standar

$-\infty \le \mu \le +\infty$, $\sigma^2 > 0$

- Merupakan bentuk standarisasi dari distribusi Normal (μ, σ^2)
- variabel random $X \sim N(\mu, \sigma^2)$ diubah menjadi $Z \sim N(0,1)$ distandarkan melalui nilai $\mu = 0$ dan $\sigma^2 = 1$.
- Fungsi peluang dist. normal baku:

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(z)^2}$$

dimana $z = \frac{x-\mu}{\sigma}$

- Mencari nilai peluang dist. Normal melalui nilai standarisasi z

1. ubah nilai x menjadi z

$$z = \frac{x-\mu}{\sigma}$$

2. z hanya boleh 2 desimal

3. $P(X=x) = P(z=z)$
   $P(X<x) = P(X \le x) = P(z<z) = P(z \le z)$] langsung lihat tabel z
   $P(X>x) = P(z>z) = 1 - P(z \le z)$
   $P(X \ge x) = P(z \ge z) = 1 - P(z < z)$] lihat nilai peluang di tabel z

$= 1 - P(X \le 18000)$

$= 1 - \int_{-\infty}^{18000} \frac{1}{} e^{-\frac{1}{2}\left(\frac{x-15015}{3540}\right)^2} dx$

$X \sim N(15015, 12536$

DISTRIBUSI NORMAL Baku / standar

- Merupakan bentuk standarisasi dari distribusi Normal (μ, σ^2)
- variabel random $X \sim N(\mu, \sigma^2)$ diubah menjadi $Z \sim N(0,1)$ distandarkan melalui nilai $\mu = 0$ dan $\sigma^2 = 1$.
- Fungsi peluang dist. normal baku:

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(z)^2}$$

dimana $z = \frac{x-\mu}{\sigma}$

- Mencari nilai peluang dist. Normal melalui nilai standarisasi z

1. ubah nilai x menjadi z

$$z = \frac{x-\mu}{\sigma}$$

2. z hanya boleh 2 desimal

3. $P(X=x) = P(z=z)$
   $P(X<x) = P(X\leq x) = P(z<z) = P(z\leq z)$] langsung lihat tabel z
   $P(X>x) = P(z>z) = 1 - \boxed{P(z\leq z)}$
   $P(X\geq x) = P(z\geq z) = 1 - \boxed{P(z<z)}$ lihat nilai peluang di tabel z

$= 1 - P(X \leq 18000)$ $\quad X \sim N(15015, 1253600)$

$$= 1 - \int_{-\infty}^{18000} \frac{1}{} \; e^{-\frac{1}{2}\left(\frac{x-15015}{3540}\right)^2} dx$$

$X \sim N(8.46, 0.84)$

$\sigma^2 = 0.913^2 = 0.84$

$P(8 \leq X \leq 10) = P(X \leq 10) - P(X \leq 8)$

$\rightarrow P(X \leq 10) \rightarrow z = \frac{10 - 8.46}{0.913} = 1.69$

$\rightarrow$ lihat tabel : $P(z \leq 1.69) : 0.9545$

$\rightarrow P(X \leq 8) \rightarrow z = \frac{8 - 8.46}{0.913} = -0.50$

$\rightarrow$ lihat tabel : $P(z \leq -0.50) = 0.3085$

$\rightarrow P(8 \leq X \leq 10) = 0.9545 - 0.3085$

$= 0.6460$

a.) $X \sim N(8,46; 0,84)$

$P(X \geq 10) \Rightarrow z = \frac{10 - 8,46}{0,913}$

$= 1,69$

$P(X \geq 10) \Rightarrow P(z \geq 1,69)$

$= 1 - P(z < 1,69)$

$= 1 - 0,9545$

$= 0,0455$

2. $X \sim N(S$

$P(X \leq x) =$

b. $P(X < 10.000) \rightarrow$

$P(X < 10.000) = P($

c. $P(12.000 < X <$

$X = 18.000 \rightarrow z = 0,8$

$X = 12.000 \rightarrow z = \frac{12.000}{}$

$= 1 - P(x \leq 18000)$ $\quad X \sim N(15015, 12531600)$

DISTRIBUSI NORMAL Baku / standar

$-\infty \leq \mu \leq \infty$; $\sigma^2 > 0$

- merupakan bentuk standaricasi dari distribusi Normal (μ, σ^2)
- variabel random $X \sim N(\mu, \sigma^2)$ diubah menjadi $Z \sim N(0,1)$ distandarkan melalui nilai $\mu = 0$ dan $\sigma^2 = 1$. (normal baku.)
- Fungsi peluang dist. normal baku:

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(z)^2}$$

dimana $z = \frac{x - \mu}{\sigma}$

B $P(X > 15) \rightarrow Z = \frac{15 - 8,46}{0,913}$

$= 7,16$

$P(X > 15) = P(Z > 7,16)$

$= 1 - (Z \leq 7,16)$

$= 1 - 1 = 0$ //

$z_{max} = 3,49 \approx 1$

$z_{min} = -3,49 \approx 0$

$z = -5,25$

$P(z < -5,25) = 0$ //

$= 1 - P(x \leq 18000)$ $X \sim N(15015, 12531600)$

$(x - 15015)^2$

DISTRIBUSI Eksponensial

- distribusi 4/ var. random kontinyu dengan domain $x \geq 0$
- var. random X berdistribusi eksponensial dapat dituliskan

  $X \sim Exp(\lambda)$ dengan λ = rata-rata atau laju.
- Fungsi peluang :

$$f(x) = \begin{cases} \lambda e^{-\lambda . x} & ; x \geq 0 \\ 0 & ; x \text{ lainnya} \end{cases}$$

contoh :

Waktu pengerjaan ujian statistika memiliki rata-rata waktu 90 menit. Hitunglah waktu seorang mahasiswa mengerjakan ujian dalam waktu :

1. 88 Menit
2. kurang dari 100 menit
3. lebih dari 75 menit.

$z_{max} = 3,49 \approx 1$

$z_{min} = -3,49 \approx 0$

$z = -5,25$

$P(z < -5,25) = 0$

17. $X \sim N(15.015, 3540^2)$

a. $P(X > 18.000) \rightarrow z = \frac{x-\mu}{\sigma} = \frac{18.000 - 15.015}{3540} = 0,84$

$P(X > 18.000) = P(z > 0,84)$

$= 1 - P(z \leq 0,84) =$

$= 1 - 0,7995 \rightarrow$ tabel z

$= 0,2005 = 0,2$ //

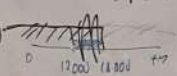
2. $X \sim N(500, 64)$

$P(X \leq x) = \int_{-\infty}^{540} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$

b. $P(X < 10.000) \rightarrow z = \frac{10.000 - 15.015}{3540} = -1,42$

$P(X < 10.000) = P(z < -1,42) = 0,0778 \approx 0,078$ //

c. $P(12.000 < X < 18.000)$



$X = 18.000 \rightarrow z = 0,84$

$X = 12.000 \rightarrow z = \frac{12.000 - 15.015}{3540} = -0,85$

$X \leq 18.000 -$

$X \leq 12.000$

$P(12.000 < X < 18.000) = P(X \leq 18.000) - P(X \leq 12.000)$

$= P(z \leq 0,84) - P(z < -0,85)$

$= 0,7995 - 0,1977$

$= 0,6018$

$= 1 - P(x \leq 18000)$ $X \sim N(15015, 1253600)$

$(x - 15015)^2$

DISTRIBUSI Eksponensial

- distribusi y/ var. random kontinyu dengan domain $X \geq 0$
- var. random X berdistribusi eksponensial dapat dituliskan

  $X \sim Exp(\lambda)$ dengan λ : rate / laju.
- Fungsi peluang :

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & ; x \geq 0 \\ 0 & ; x \text{ lainnya} \end{cases}$$

$$E(x) = \mu = \frac{1}{\lambda}$$

$90 = \frac{1}{\lambda}$

$\lambda = 1/90$

contoh :

Waktu pengerjaan ujian statistika memiliki rata-rata waktu 90 menit. Hitunglah waktu seorang mahasiswa mengerjakan ujian dalam waktu :

1. 88 menit $= P(x = 88) = f(88)$
2. kurang dari 100 menit $P(x \leq 100) = F(100)$
3. lebih dari 75 menit. $P(x > 75) = 1 - P(x \leq 75$
   $= 1 - F(75)$

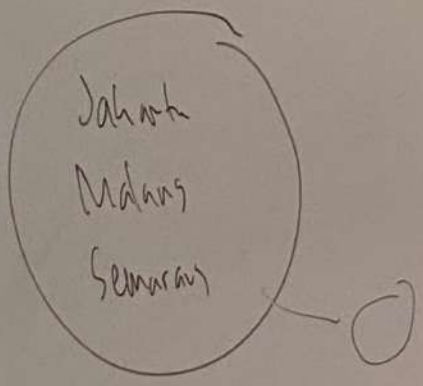
$z_{max} = 3,49 \approx 1$

$z_{min} = -3,49 \approx 0$

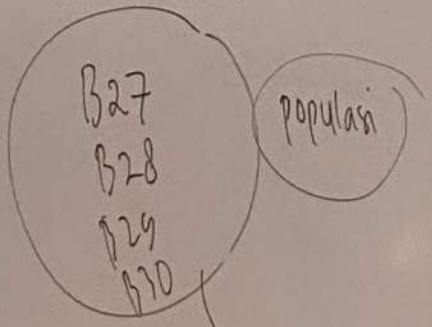
$z = -5,25$

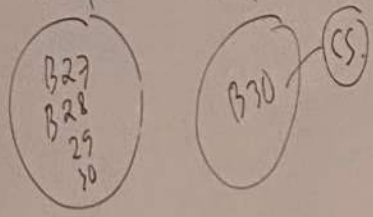
$P(z < -5,25) = 0$

Binus University

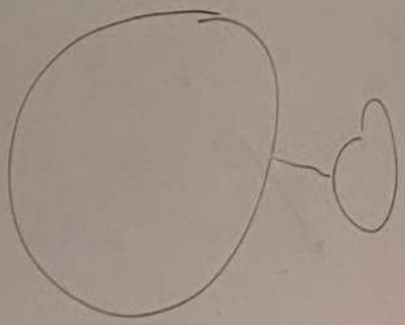


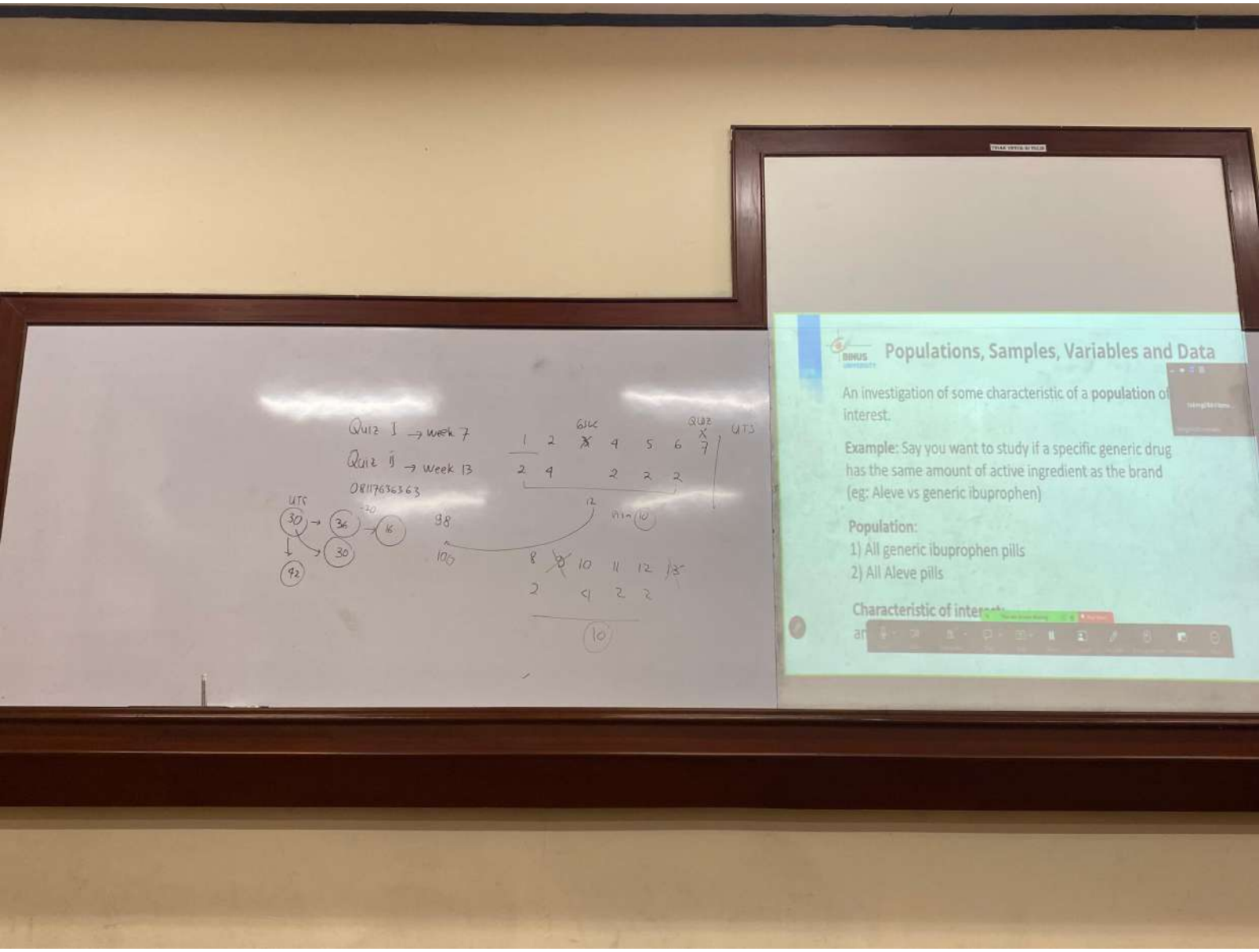
Compt. Engineering

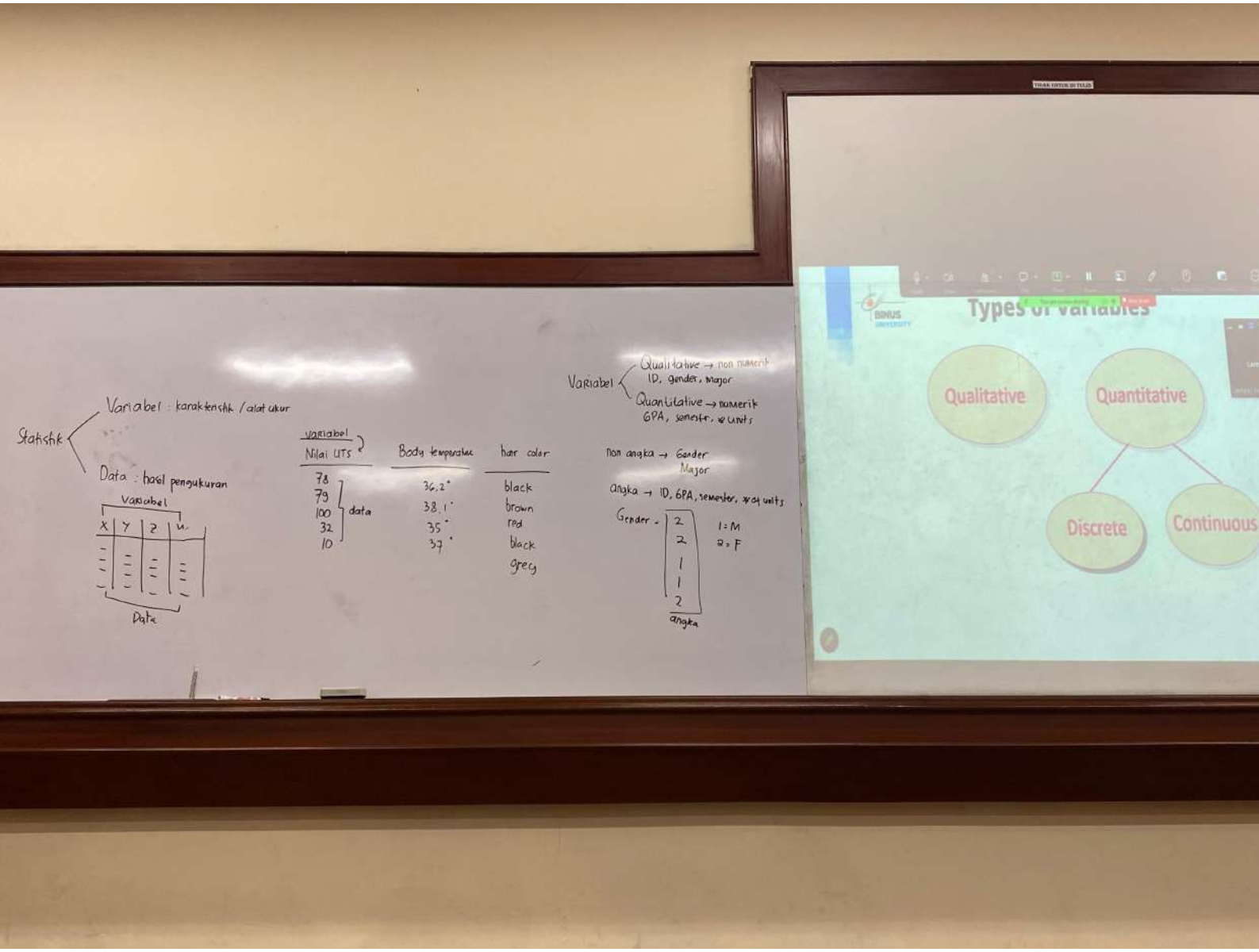


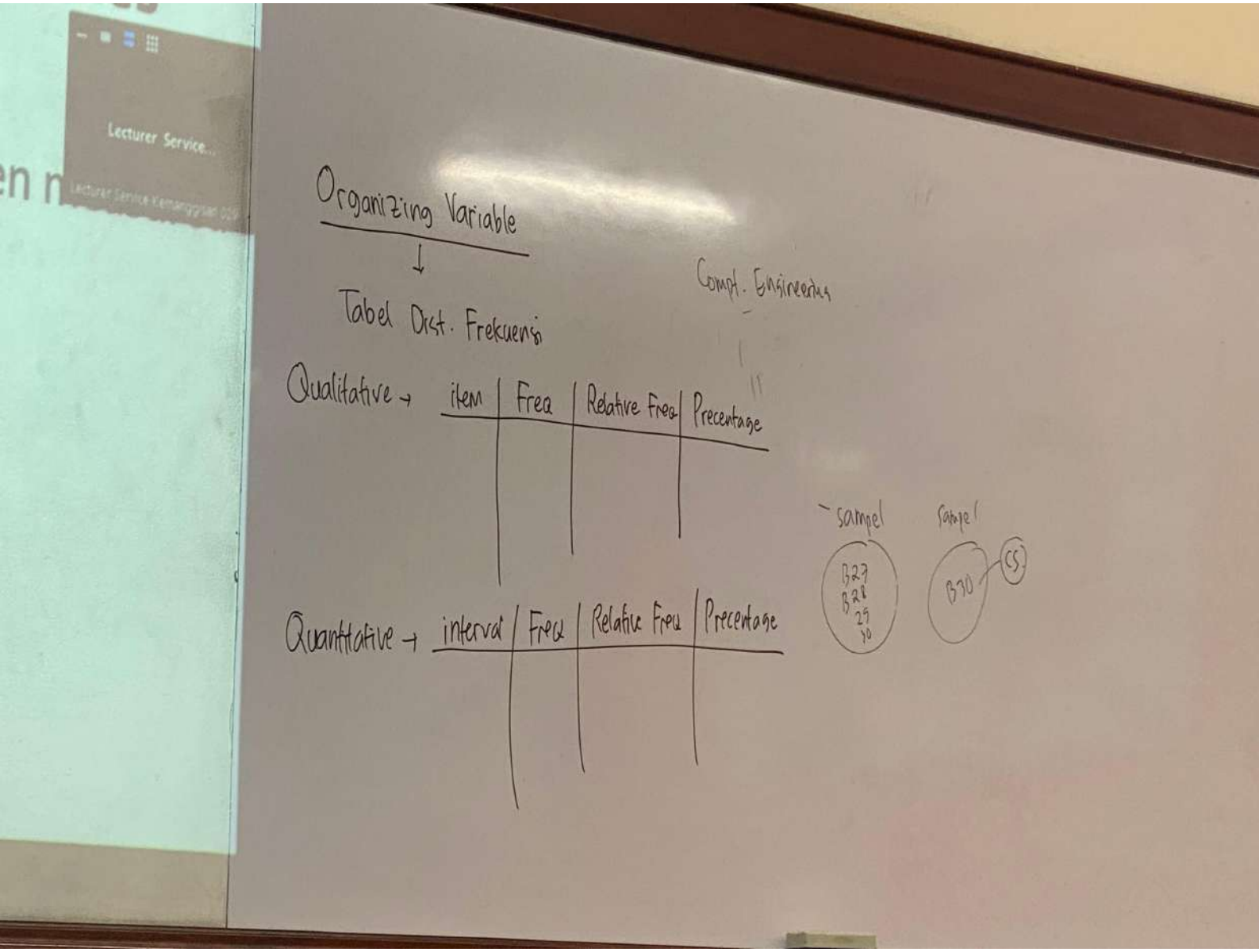


BINUS greater Thf




Quiz 1 → week 7
Quiz 2 → week 13
08117636363
UTS
30
36
16
30
92
98
100
Kuis
QUIZ
X
UTS
1 2 4 5 6 7
2 4 2 2 2
12
min 10
8 9 10 11 12 13
2 <1 2 2
10
Populations, Samples, Variables and Data
An investigation of some characteristic of a population of interest.
Example: Say you want to study if a specific generic drug has the same amount of active ingredient as the brand (eg: Aleve vs generic ibuprophen)
Population:
1) All generic ibuprophen pills
2) All Aleve pills
Characteristic of interest:


Statistik
Variabel : karakteristik / alat ukur
Data : hasil pengukuran
Variabel
X Y Z U
Data
Variabel
Nilai UTS
78
79
100
32
10
data
Body temperature
36,2°
38,1°
35°
37°
hair color
black
brown
red
black
grey
Variabel
Qualitative → non numerik
ID, gender, major
Quantitative → numerik
GPA, semester, # units
non angka → Gender
Major
angka → ID, GPA, semester, # of units
Gender
2
2
1
1
2
1 = M
2 = F
angka
Types of variables
Qualitative
Quantitative
Discrete
Continuous


Lecturer Service...
Organizing Variable
Tabel Dist. Frekuensi
Compt. Engineering
Qualitative → item | Freq | Relative Freq | Precentage
Quantitative → interval | Freq | Relative Freq | Precentage
sampel
sampel
B27
B28
25
30
B30
CS

Lecturer Service...

Organizing Variable

↓

continuous data

Const. Engineering

17,23 18,57 18,78 18,79 20,10 21,31 23,49
17,2 18,6 18,8 18,8 20,1 21,3 23,5

stem	leaf
17	2
18	6 8 8
19	
20	1
21	3
22	5
23	.

(.)

Stem and leaf (numerical data)

1. urutkan kecil → besar
2. pisahkan satuan dan non satuan

40 60 65 68 70 ... 95 123 10573

stem (puluhan)	leaf (satuan)
4	0
5	
6	0 5 5 5 8 8
7	0
8	
9	0 5
10	
11	
12	3
⋮	⋮
1057	3